

KADI SARVA VISHWAVIDHYALAYA

B.E. Semester VII EXAMINATION (March - 2023)

Subject Code: CT704A-N
Date: 29/03/2023

Time: 10.00 am to 1.00 pm

Subject Name: Distributed Systems
Total Marks: 70 Marks

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Use of scientific calculator is permitted.
4. Indicate clearly, the options you attempt along with its respective question number.
5. Use the last page of main supplementary for rough work.

Section-I

Q:1 Attempt Following.

- (A) Explain client server model with diagram. [5]
- (B) List and explain different types of networks. [5]
- (C) Differentiate between distributed systems and centralize system. [5]

OR

- Q:1 (C) Define WWW and describe working of WWW. [5]

Q:2 Answer the following question.

- (A) Explain IPC message format. [5]
- (B) Write brief note on ATM protocol. [5]

OR

- Q:2 (A) List any five use cases of distributed system. [5]
- (B) Define distributed system and write advantages and disadvantages of DS. [5]

Q:3 Answer the following question.

- (A) Describe IPC synchronization in detail. [5]
- (B) Why clock synchronization is required in distributed system? [5]

OR

- Q:3 (A) Write note on RMI architecture. [5]
- (B) List and explain the need of message buffering strategies. [5]

Section II

Q:4 (All Compulsory)

- (A) Compare process vs thread. [5]
- (B) List and describe three cloud computing services. [5]
- (C) Explain service oriented architecture. [5]

OR

Q:4 (C) Explain group communication in detail. [5]

Q:5 Answer the following question.

- (A) What are the design issues in distributed shared memory system? [5]
- (B) Explain Bully algorithm with diagram. [5]

OR

Q:5 (A) Write note on distributed resource management. [5]
(B) Explain semantics File Sharing [5]

Q:6 Answer the following question.

- (A) Explain RSA algorithm in detail. [5]
- (B) Explain the requirement of Digital signature in distributed environment. [5]

OR

Q:6 (A) Describe types of File Sharing Semantic. [5]
(B) Explain strict consistency in context of distributed shared memory. [5]